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Question

In their 2022 paper, Christos Dimopoulos et al., having granted that the existence of antigravity—in which antimatter and matter repel rather than attract each other—lacked affirmative experimental support, rightly argued that such antigravity was worth considering on theoretical grounds given that evidence against it was similarly lacking. But a 2023 report by an international team of researchers details the first direct ballistic observations of antihydrogen atoms under gravity inside a CERN particle accelerator. Corresponding most closely to predictions under gravitational attraction, these observations were thoroughly inconsistent with antigravity.

Which choice best states the main idea of the text?

Answer

- A. Antihydrogen ballistics observations were conducted at CERN to test specific conclusions about antigravity presented in the 2022 paper by Dimopoulos et al.
- B. Although theoreticians were justified in studying antigravity before the release of the 2023 report, the report's findings suggest that the rationale for theoretical consideration offered in the 2022 paper by Dimopoulos et al. is no longer applicable.
- C. The theoretical approach represented in the 2022 paper by Dimopoulos et al. assumed that unambiguous proof of antigravity would not be achievable, but the results in the 2023 report undermine that assumption.
- D. Before 2023, researchers' inordinate focus on theoretical considerations hindered the development of the experimental regimen for direct antihydrogen ballistics observations.